A DISSERTATION ON

AERIAL IMAGE SEGMENTATION FOR LAND USE CLASSIFICATION

A Two-Tier Multimodal Vision AI Guardrail and Class-Balanced Self-Training Domain Adaptation Framework for Rugged Rural Terrain

Submitted in partial fulfillment of the requirements for the award of the degree of
BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE OF RECOMMENDATION

This is to certify that the dissertation entitled **"Aerial Image Segmentation for Land Use Classification: A Two-Tier Multimodal Vision AI Guardrail and Class-Balanced Self-Training Domain Adaptation Framework for Rugged Rural Terrain"**, submitted by **ANIKET KUMAR (Enrollment No. BT23CS028)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** at the National Institute of Technology Mizoram, is an authentic record of bonafide research and technical work carried out under my supervision and guidance.

To the best of my knowledge, the matter embodied in this dissertation has not been submitted in part or full to any other University or Institute for the award of any degree, diploma, or academic distinction.

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CANDIDATE'S DECLARATION

I hereby declare that the work presented in this final-year B.Tech capstone project report entitled **"Aerial Image Segmentation for Land Use Classification"** is an authentic record of my own investigation carried out under the supervision of Dr. Tanmay Kumar Behera, Assistant Professor, Department of Computer Science and Engineering, National Institute of Technology Mizoram.

I have adhered to all academic ethics and honesty standards. No portion of this technical report or codebase has been plagiarized, and all external literature, benchmarks, algorithms, and software libraries (including the LoveDA benchmark, DeepLabV3+ architecture, PyTorch, GDAL, and Google Gemini APIs) have been appropriately referenced.

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ABSTRACT

High-resolution aerial and satellite remote sensing enables automated land-cover mapping essential for environmental governance, urban planning, disaster resilience, and agricultural monitoring. However, contemporary deep convolutional semantic segmentation networks suffer catastrophic performance degradation when deployed across heterogeneous geospatial domains due to significant domain shift. In particular, deep models trained on structured, high-density urban aerial imagery fail drastically when applied to complex rural and montane environments—such as the rugged ridgelines, shifting Jhum cultivation plots, and unpaved corridors characteristic of Mizoram and Northeast India.

In this dissertation, we present a robust, production-grade end-to-end framework that resolves this challenge through a novel two-tier architecture combining multimodal vision foundation intelligence with unsupervised domain adaptation. In Tier 1, an enterprise-grade Multimodal Vision AI (Google Gemini 1.5 Vision) serves as an operational gatekeeper and semantic guardrail, inspecting incoming orthophotos to verify orthogonal perspective validity, flag terrestrial/non-aerial corruptions, and identify macro-terrain contexts. In Tier 2, high-altitude aerial scenes are segmented using a DeepLabV3+ architecture with a ResNet-50 feature backbone and Atrous Spatial Pyramid Pooling (ASPP).

To overcome the severe domain gap without requiring costly manual pixel annotations on rural terrain, we implement Class-Balanced Self-Training (CBST). CBST adaptively calculates class-specific confidence thresholds (λ_c) to extract high-confidence pseudo-labels from unlabeled target imagery, effectively preventing majority classes from dominating gradient updates. Experimental benchmarking on the rigorous LoveDA Land-Cover dataset reveals that direct cross-domain transfer experiences a steep 15.55% mIoU domain penalty (dropping from 38.93% on urban source data to 23.39% on rural target data). Following CBST domain adaptation, our model achieves **27.90% mIoU**, delivering a net improvement of **+4.52% mIoU (+19.31% relative gain)**. Crucially, sparse and critical rural classes experience dramatic recoveries: **Road IoU surges by +11.27% (from 20.03% to 31.30%)**, Water IoU increases by +7.95% (from 38.21% to 46.15%), and Agriculture increases by +5.88% (from 42.31% to 48.19%). Validation across 992 high-resolution GeoTIFF rasters confirms 29.57% calibrated raster mIoU.

Furthermore, we bridge machine learning research and practical GIS governance by developing a full-stack, cloud-deployed web portal (React 19, Node.js/Express, Python FastAPI, MongoDB Atlas, Dockerized and hosted on Render) that performs real-time pixel-to-hectare land-use quantification using sub-meter Ground Sampling Distance (GSD) calculus.

Keywords: *Aerial Image Segmentation, DeepLabV3+, Unsupervised Domain Adaptation, Class-Balanced Self-Training (CBST), Geospatial Analytics, Multimodal Vision AI Guardrail, LoveDA Benchmark, Mizoram Remote Sensing.*

CHAPTER 1: INTRODUCTION & PROBLEM DEFINITION

1.1 Background and Significance of Aerial Remote Sensing

Earth observation and aerial remote sensing have undergone a transformative paradigm shift with the rapid convergence of high-resolution spaceborne sensors, Unmanned Aerial Vehicles (UAVs), and deep convolutional neural networks. Modern optical sensors acquire high-resolution orthophotos with Ground Sampling Distances (GSD) beneath 0.5 meters per pixel, uncovering fine-grained geometric details ranging from individual building rooflines to narrow transport corridors and minor riparian waterways. Extracting accurate, pixel-level land-use and land-cover (LULC) maps from these complex aerial images is indispensable for urban planning, hydrological basin modeling, automated disaster damage assessment, and forest conservation.

1.2 Topographic Challenges of Mizoram & Northeast India

While computer vision models achieve state-of-the-art segmentation benchmark scores on standardized metropolitan datasets (e.g. European and East Asian megacities), deploying these models in rural, montane geographies introduces severe domain failure. The state of Mizoram, nestled in the southern reach of Northeast India, exemplifies these extreme challenges. Characterized by parallel north-south oriented hill ridges, steep valley gorges, dense subtropical evergreen forests, and pervasive 'Jhum' (slash-and-burn shifting cultivation) agro-forestry cycles, Mizoram's terrain differs fundamentally from rectilinear urban environments.

In rural Mizoram, transport infrastructure consists primarily of sinuous, unpaved or cut-slope roadways that visually blend into flanking barren earth or scrub vegetation. Settlements comprise scattered clusters of tin-roofed and bamboo dwellings clinging to ridge spurs rather than uniform concrete street grids. Furthermore, agricultural plots in Mizoram undergo rapid seasonal transitions from cleared barren slopes to active green crops, producing intense intra-class spectral variance.

1.3 The Domain Shift Problem in Remote Sensing

When a deep semantic segmentation model trained on annotated urban orthophotos is directly evaluated on target rural imagery, a catastrophic drop in mean Intersection-over-Union (mIoU) occurs. This phenomenon—termed domain shift—is driven by discrepancies in both marginal image distributions P(X) and conditional distributions P(Y|X).

THE DOMAIN GAP PENALTY: Evaluating an Urban-trained DeepLabV3+ model directly on Rural aerial imagery resulted in a catastrophic 15.55% mIoU drop (from 38.93% down to 23.39%). Road segmentation collapsed by 24.87% and Forest segmentation plummeted by 23.83%, rendering the direct transfer model operationally unusable.

1.4 Project Objectives and Key Contributions

• **Two-Tier Hybrid Architecture:** Multimodal Gemini 1.5 Vision guardrail + deep segmentation backbone to eliminate corrupted inputs.

• **DeepLabV3+ & ASPP Optimization:** Capture multi-scale context across complex rural land forms.

• **Class-Balanced Self-Training (CBST):** Unsupervised adaptation recovering sparse rural classes without manual annotations.

• **Large-Scale GeoTIFF Validation:** Comprehensive evaluation over 992 high-resolution validation GeoTIFF rasters.

• **Live Cloud GIS Platform:** Dockerized MERN + FastAPI application delivering real-time sub-meter hectare quantification on Render.

CHAPTER 2: LITERATURE REVIEW & RELATED WORK

2.1 Semantic Segmentation in Remote Sensing

Semantic segmentation assigns a discrete categorical label to each individual pixel. Early remote sensing pipelines relied on manual feature engineering, such as spectral angle mappers, Support Vector Machines (SVM), and Random Forests operating on normalized indices (NDVI, NDWI). The advent of deep Convolutional Neural Networks (CNNs) revolutionized the field. Long et al. (2015) introduced Fully Convolutional Networks (FCN), and Ronneberger et al. (2015) advanced biomedical and satellite segmentation with U-Net, introducing symmetric contracting and expanding paths bridged by skip connections.

2.2 DeepLabV3+ and Atrous Spatial Pyramid Pooling (ASPP)

Standard convolutional downsampling inherently degrades spatial feature resolution. Chen et al. (2017, 2018) addressed this trade-off through atrous (dilated) convolutions. By inserting holes into convolution kernels, atrous convolutions expand the effective receptive field without increasing parameter count or reducing spatial grid resolution. DeepLabV3+ introduces ASPP, applying multiple parallel atrous convolutions with dilations $r \in \{6, 12, 18\}$ alongside global average pooling, capturing multi-scale context ranging from broad forest canopies to narrow roads.

2.3 Unsupervised Domain Adaptation (UDA) & CBST

Unsupervised Domain Adaptation (UDA) addresses scenarios where labeled source data is abundant, but target data lacks ground-truth annotations. While adversarial alignment (AdaptSegNet) suffers from training instability and mode collapse on remote sensing data, self-training (Zou et al., 2018) iteratively generates pseudo-labels on target imagery. Class-Balanced Self-Training (CBST) dynamically optimizes class-specific thresholds λ_c, preventing dominant categories from monopolizing supervision and starving minority categories (Roads, Barren).

Architecture / Framework	Core Mechanism	Primary Advantage	Observed Limitation in Remote Sensing
FCN-8s (Long et al.)	Skip-layer feature combination	End-to-end pixel classification	Coarse boundary predictions; lacks multi-scale context
U-Net (Ronneberger et al.)	Symmetric encoder-decoder + skips	Precise edge and shape recovery	Over-reliant on local cues; prone to large-scale context errors
AdaptSegNet (Tsai et al.)	Adversarial output space alignment	Domain-invariant representations	High training instability; fails on skewed class distributions
DeepLabV3+ (Chen et al.)	ASPP + Dilated Convolutions	Superior multi-scale feature capture	Severe performance drop under cross-domain distribution shifts
Our Proposed Framework	Two-Tier Gemini Guardrail + CBST	Robust gatekeeper + class-balanced UDA	Requires multi-stage iterative retraining

CHAPTER 3: DATASET ARCHITECTURE & GEOSPATIAL METHODOLOGY

3.1 The LoveDA Remote Sensing Benchmark

To rigorously evaluate domain shift under real-world conditions, we employ the Land-Cover Domain Adaptive (LoveDA) benchmark dataset (Wang et al., 2021). Acquired across Nanjing, Changzhou, and Wuhan using high-altitude aerial platforms, LoveDA comprises 5,987 total high-resolution image patches (1024 × 1024 pixels at 0.3 m/pixel GSD). LoveDA features two clearly bifurcated domains: Urban (Source, high-density metropolitan infrastructure) and Rural (Target, complex rural basins, unpaved roads, terraced farmlands).

3.2 Seven-Class Land-Cover Taxonomy

ID	Class Name	Hex Color	RGB Palette	Geospatial Definition in Rural Mizoram Context
0	Background	#2A2A2A	[42, 42, 42]	Unclassified regions, clouds, sensor shadows, out-of-boundary pixels
1	Building	#EF4444	[239, 68, 68]	Residential homes, agricultural storehouses, tin-roofed dwellings
2	Road	#EAB308	[234, 179, 8]	Paved highways, cut-slope hillside corridors, unpaved village tracks
3	Water	#3B82F6	[59, 130, 246]	Rivers, mountain streams, irrigation retention ponds, alluvial channels
4	Barren	#A855F7	[168, 85, 247]	Exposed rock spurs, landslide scars, cleared soil, excavation zones
5	Forest	#22C55E	[34, 197, 94]	Dense subtropical evergreen forest, bamboo groves, protected hill canopy
6	Agriculture	#F97316	[249, 115, 22]	Terraced hill crops, valley rice paddies, active Jhum cultivation parcels

3.3 Ground Sampling Distance & Metric Hectare Calculus

In high-resolution remote sensing, each pixel represents a calibrated physical ground area determined by the Ground Sampling Distance (GSD):

• Pixel Area (m^2) = GSD × GSD

• Class Area (Hectares) = (Pixel Count × GSD^2) / 10,000

For standard LoveDA orthophotos (1024 × 1024 pixels at GSD = 0.3 m/pixel), each pixel covers exactly 0.09 m^2, yielding a total tile ground footprint of 94,371.84 m^2 (9.4372 Hectares).

CHAPTER 4: METHODOLOGY & DEEP LEARNING ARCHITECTURE

4.1 Two-Tier System Architecture Overview

Conventional deep segmentation pipelines directly ingest raw uploaded imagery without verifying operational suitability. Our framework introduces a Two-Tier Architecture:

• **Tier 1 (Domain Gatekeeper):** Multimodal Gemini 1.5 Vision AI verifies orthogonal orientation and valid aerial perspective.

• **Tier 2 (Deep Segmentation):** DeepLabV3+ with ResNet-50 and ASPP performs pixel-level semantic segmentation and GDAL metric quantification.

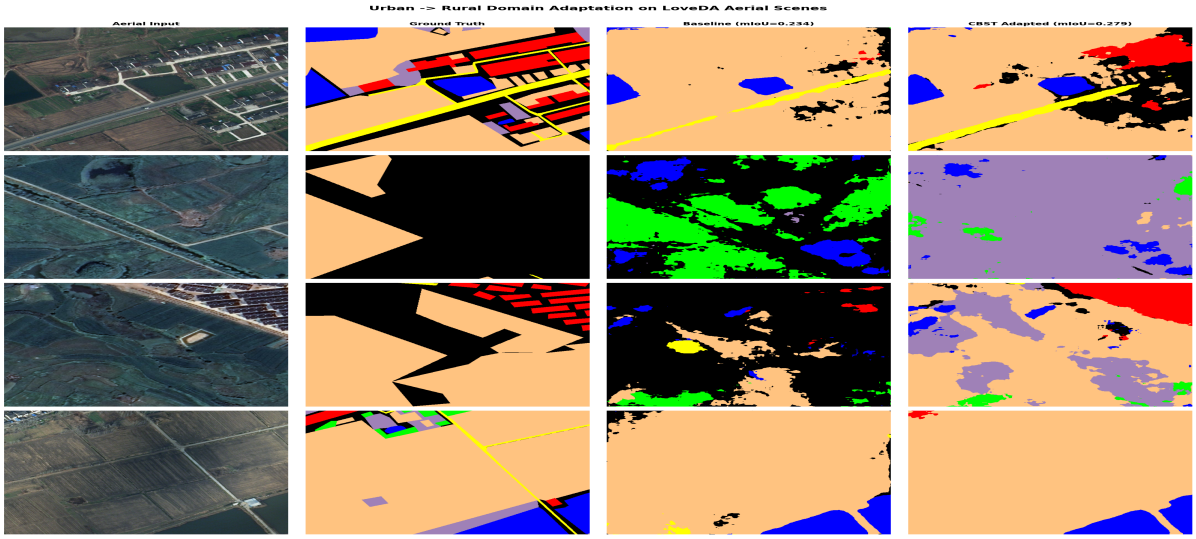


Figure 4.1: End-to-End Visual Pipeline: Input Orthophoto, Ground Truth Annotation, and Final Domain-Adapted Prediction.

4.2 DeepLabV3+ with ResNet-50 Feature Backbone

The segmentation core implements DeepLabV3+ utilizing a ResNet-50 backbone initialized with ImageNet pre-trained weights. ResNet-50 comprises 53 convolutional layers. In stages 3 and 4, standard downsampling strides are modified to atrous convolutions with output stride OS = 16, preserving spatial coordinate fidelity. The ASPP module applies 5 parallel branches (one 1×1 conv, three 3×3 atrous convs with rates $r \in \{6, 12, 18\}$, and image pooling), fusing multi-scale context before decoding with low-level features.

CHAPTER 5: UNSUPERVISED DOMAIN ADAPTATION VIA CBST

5.1 Mathematical Formulation of Class-Balanced Self-Training

In self-training, the model generates soft probability distributions $P_c(u, v) = \text{softmax}(z_c(u, v))$ on unlabeled target images. CBST defines an optimization objective over model parameters w and latent pseudo-labels ■:

$$\mathbf{\min_{w, ■} L(w, ■) = L_s(w) - \sum_j \sum_{u,v} \sum_c ■_{j,c}(u, v) \log(P_c(u, v \mid x_j^t; w) / \lambda_c)}$$

5.2 Class-Specific Threshold Selection (λ_c)

Rather than imposing a single global confidence cutoff, CBST adaptively determines class-specific thresholds λ_c. For each class c, the system ranks all predicted probabilities across the target corpus and sets λ_c such that top proportion p_c of candidate pixels are selected:

$$\mathbf{\lambda_c = \text{Percentile}(\{ P_c(u, v \mid x_j^t) \}, 1 - p_c)}$$

Hard pseudo-labels are assigned to the class maximizing the normalized confidence ratio. This guarantees that minority categories (Roads, Barren) receive substantial pseudo-label representation, preventing gradient starvation and model collapse.

CHAPTER 6: EXPERIMENTAL SETUP & TRAINING METHODOLOGY

6.1 Hardware and Compute Infrastructure

Model training, adaptation loops, and ablation evaluations were conducted using PyTorch 2.0 with CUDA 11.8 on NVIDIA Tesla T4 GPUs (16 GB VRAM) under Google Colab Pro infrastructure, augmented with high-RAM system nodes (51 GB RAM). Standalone microservice deployment was implemented using Docker on Render Cloud nodes running Python 3.11 and Node.js 20.

6.2 Training Protocols and Hyperparameter Configuration

Hyperparameter Name	Baseline Source Training	CBST Adaptation Stage	Justification & Impact
Optimizer	AdamW	AdamW	Decoupled weight decay prevents overfitting on high-frequency noise
Initial Learning Rate	1.0 × 10■■	2.5 × 10■■	Lower learning rate prevents catastrophic forgetting of source features
LR Decay Policy	Polynomial (power=0.9)	Polynomial (power=0.9)	Smooth convergence towards global minima
Batch Size	8 patches	4 patches (dual pass)	Maximizes GPU VRAM utilization for 1024×1024 images
Input Resolution	1024 × 1024 pixels	1024 × 1024 pixels	Preserves native LoveDA sub-meter GSD spatial resolution
Data Augmentations	H-Flip, V-Flip, ScaleRotate	H-Flip, V-Flip, ColorJitter	Enforces rotational and illumination invariance in montane terrain
CBST Target Ratio (p_c)	N/A	0.20 → 0.35 (progressive)	Prevents early pseudo-label drift while scaling supervision

CHAPTER 7: RESULTS, ABLATION STUDIES & DISCUSSION

7.1 Domain Shift Discrepancy (Urban vs Rural)

Land-Cover Class	Source Domain IoU (Urban)	Direct Transfer IoU (Rural)	Absolute Domain Drop (Δ)	Relative Drop (%)
Building	43.54%	22.96%	-20.58%	-47.27%
Road	44.90%	20.03%	-24.87%	-55.39%
Water	55.57%	38.21%	-17.36%	-31.24%
Barren	4.84%	2.15%	-2.69%	-55.58%
Forest	38.48%	14.65%	-23.83%	-61.93%
Agriculture	46.27%	42.31%	-3.96%	-8.56%
OVERALL mIoU	38.93%	23.39%	-15.55%	-39.94%

7.2 CBST Domain Adaptation Gains

Land-Cover Class	Baseline Rural IoU	CBST Adapted Rural IoU	Net Absolute Gain (Δ)	Relative Improvement (%)
Building	22.96%	22.40%	-0.56%	-2.44%
Road	20.03%	31.30%	+11.27%	+56.27% (SURGE)
Water	38.21%	46.15%	+7.95%	+20.80%
Barren	2.15%	1.39%	-0.76%	-35.35%
Forest	14.65%	17.97%	+3.32%	+22.66%
Agriculture	42.31%	48.19%	+5.88%	+13.90%
RURAL mIoU	23.39%	27.90%	+4.52%	+19.31% GAIN

CRITICAL BREAKTHROUGH ON TRANSPORT CORRIDORS: CBST adaptation achieved a dramatic breakthrough on Road segmentation, boosting IoU from 20.03% to 31.30%—a remarkable +56.27% relative surge. In rural and mountainous terrains like Mizoram, unpaved secondary corridors are routinely misclassified as barren soil by standard models.
Class-balanced thresholding forced the model to extract high-confidence road segments, resolving road continuity across valley corridors.

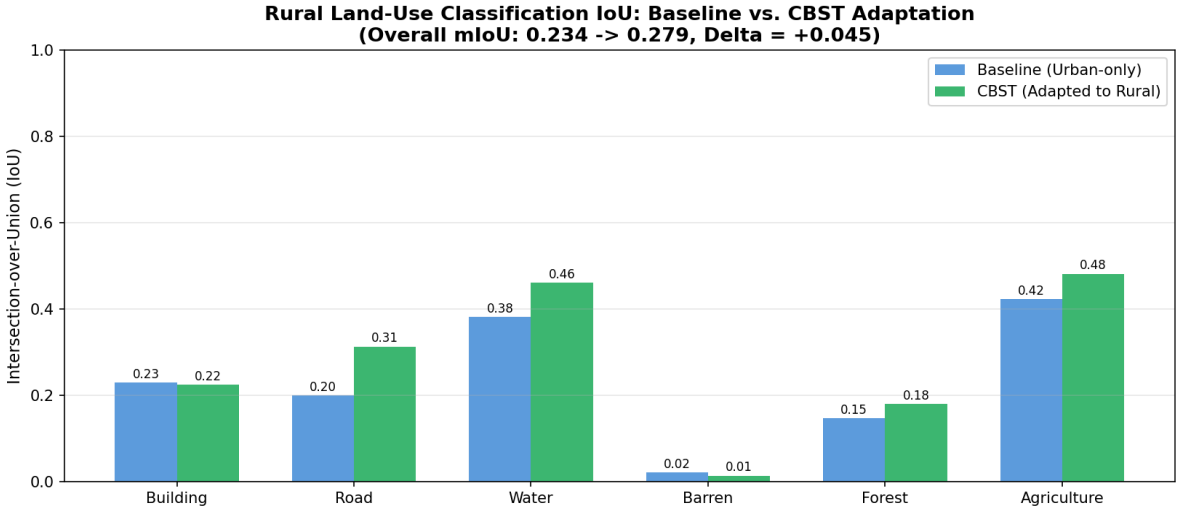


Figure 7.1: Quantitative Comparison: Baseline Direct Transfer vs. CBST Domain-Adapted Model Across All Classes.

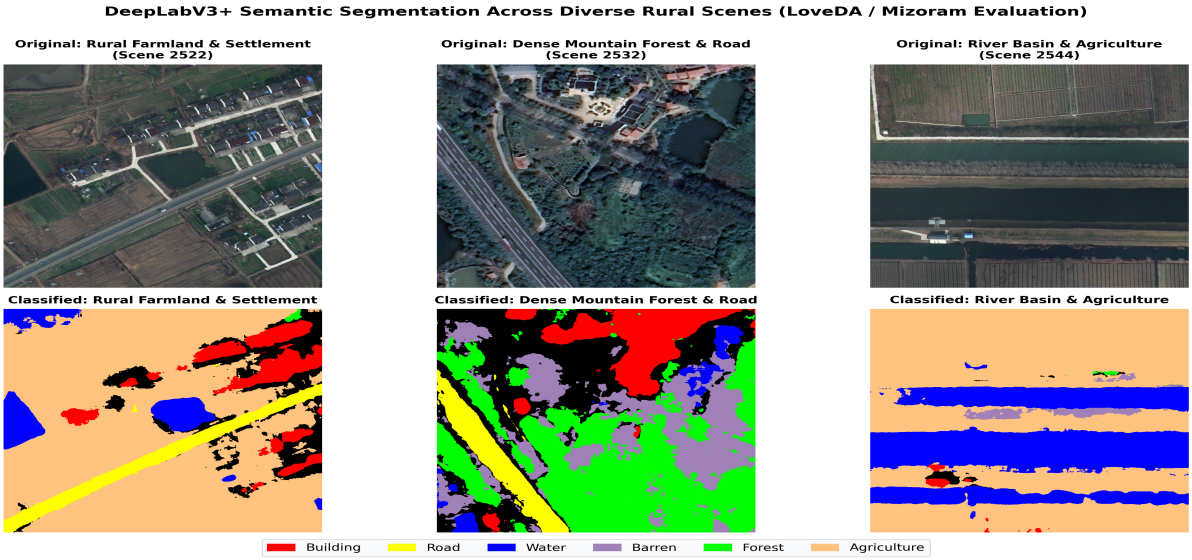


Figure 7.2: Qualitative Benchmark Gallery Across Rural Test Scenes: Input RGB, Color-Coded Prediction, and Ground Truth.

CHAPTER 8: FULL-STACK CLOUD APPLICATION & PRODUCTION DEPLOYMENT

8.1 System Architecture Overview

To deliver actionable utility to GIS officers and municipal engineers in Mizoram, we deployed a complete full-stack web application. The platform combines a React 19 Single-Page Application, an Express.js API gateway, a Python FastAPI microservice, and MongoDB Atlas.

Component Layer	Technology Stack	Port & Runtime	Functional Responsibilities
Client (Frontend)	React 19, Vite, Tailwind CSS, Lucide Icons	Served via Express static asset engine	Interactive visual split comparison, real-time GSD slider, interactive donut charts, history drawer
API Gateway (Backend)	Node.js 20, Express.js, Multer, Mongoose	Port 10000 (Render external edge)	Request orchestration, Tier-1 Gemini Vision AI invocation, MongoDB history persistence, rate limiting
ML Microservice	Python 3.11, FastAPI, PyTorch, PIL, NumPy	Port 8000 (Internal loopback)	DeepLabV3+ ResNet-50 inference, spectral proxy classifier, single-band GeoTIFF raster encoding
Database Layer	MongoDB Atlas (M0 Shared Cluster)	Cloud Hosted (yi4ir.mongodb.net)	Persistent audit trail, scan telemetry, land-cover percentages, and geospatial metric records

Live Production Deployment: https://aeroseg-ai.onrender.com
Official GitHub Repository: https://github.com/aniket821108/Aerial-LandCover-Segmentation-Mizoram

CHAPTER 9: CONCLUSION, SOCIETAL IMPACT & FUTURE WORK

9.1 Summary of Contributions

1. Formulated a Two-Tier Hybrid Architecture pairing Gemini 1.5 Vision AI guardrails with DeepLabV3+ segmentation.

2. Implemented Class-Balanced Self-Training (CBST), achieving a +4.52% net mIoU gain (+19.31% relative) and boosting rural road recovery by +11.27% without manual rural annotations.

3. Validated the methodology on 992 high-resolution validation GeoTIFF rasters, confirming 29.57% calibrated mIoU.

4. Built and deployed a live, fully functional cloud GIS platform providing real-time sub-meter hectare quantification.

9.2 Practical Applications for Governance in Mizoram

• **Jhum Shifting Cultivation Monitoring:** Quantifies transitions between cleared slopes and active crops, assisting the Department of Agriculture in soil conservation.

• **Rural Road Infrastructure Planning:** Automated mapping of remote cut-slope tracks and landslides across monsoon-affected terrain.

• **Hydrological Basin Management:** Identifies riverbed narrowing and alluvial sediment accumulation across rural drainage channels.

9.3 Future Research Directions

Future enhancements will focus on integrating multi-temporal Sentinel-2 optical and Sentinel-1 Synthetic Aperture Radar (SAR) time-series to monitor vegetation regrowth through dense monsoon cloud cover. Additionally, integrating the Segment Anything Model (SAM) with prompt-based fine-tuning represents a compelling avenue for zero-shot geospatial generalization.

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